UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. MIDI files store digitized audio samples.
2. Hypermedia combines multimedia with hyperlinks.
3. A graphics tablet is used to draw directly into a computer.
4. Keyframes define important positions in an animation.
5. A linear presentation allows users to choose their own path.
6. A multimedia project always uses every media element.
7. Multimedia combines text, graphics, audio, video and animation.
8. Frame rate is the number of video frames displayed per second.
9. Tweening creates intermediate frames between keyframes.
10. Cel animation is created by drawing each frame by hand.
11. Interlacing displays every line of a video frame at the same time.
12. CMYK is used for printed output.
13. Authoring tools help developers organise and edit multimedia elements.
14. Streaming allows media to begin playing before the whole file is downloaded.
15. Higher resolution always means a smaller file.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Keyframe	B. MIDI	C. Linear multimedia
D. Tweening	E. Sampling	F. Multimedia
G. Amplitude	H. Copyright	I. Bitmap
J. Plotter	K. Storyboard	L. HTML
M. Card-based tool	N. Scanner	O. Browser

1. integration of text, graphics, audio, video and animation
2. image made of pixels
3. converting sound into digital values
4. visual sequence plan
5. standard for musical notes and instructions
6. important frame in an animation
7. input device that digitises documents
8. software that displays web pages
9. height of a sound wave
10. language for web structure
11. generating in-between frames
12. presentation without user control of sequence
13. exclusive rights of a creator
14. output device for large drawings
15. authoring tool organised like pages or cards

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PART C (30 MARKS)

Choose the correct answer for each question.

1. A browser is used to:
   A. Print documents only
   B. Scan images
   C. Display web pages
   D. Edit audio

2. A prototype is:
   A. A final release
   B. A font
   C. An early model of a product
   D. A codec

3. Which colour model is subtractive?
   A. HSV
   B. CMYK
   C. HSL
   D. RGB

4. Kerning adjusts:
   A. Page size
   B. Space between individual character pairs
   C. Font colour
   D. Line length

5. Virtual reality creates:
   A. A text font
   B. A camera lens
   C. A printed page
   D. A simulated environment

6. Which is an output device?
   A. Plotter
   B. Scanner
   C. Keyboard
   D. Mouse

7. A hyperlink links to:
   A. Only audio
   B. Another resource
   C. Only text
   D. Only images

8. Chroma key is used to:
   A. Scan images
   B. Add text
   C. Compress audio
   D. Replace a colour background

9. Hypertext contains:
   A. Only images
   B. Links to other content
   C. Only audio
   D. No text

10. Amplitude of a sound wave relates to:
    A. Loudness
    B. Colour
    C. Frame rate
    D. Pitch

11. Which is an input device?
    A. Monitor
    B. Scanner
    C. Plotter
    D. Printer

12. A wireframe is:
    A. A codec
    B. A low-fidelity layout sketch
    C. A final video
    D. A sound file

13. Interlacing:
    A. Adds sound
    B. Removes colour
    C. Draws every line at once
    D. Draws alternate lines of a frame

14. Which is an authoring tool category?
A. Card-based
B. RGB
C. JPEG
D. CMYK

15. An authoring tool is used to:
A. Only browse the web
B. Only print text
C. Assemble multimedia elements
D. Only record audio

16. Which best defines multimedia?
A. A single image
B. A printed book
C. Text, graphics, audio, video and animation
D. Text only

17. RAM is:
A. A web browser
B. An input device
C. Permanent storage
D. Temporary working memory

18. A storyboard is:
A. A sound file
B. A colour model
C. A visual sequence plan
D. A video codec

19. CMYK is used mainly for:
A. Printed output
B. Sound mixing
C. Screen display
D. Animation

20. Sampling converts:
A. Images to vectors
B. Text to speech
C. Digital video to text
D. Analog sound to digital values

21. A bitmap image is made of:
A. Mathematical objects
B. Sound samples
C. Text characters
D. Pixels

22. A serif font has:
A. Small strokes at character ends
B. No spacing
C. Only uppercase letters
D. No strokes

23. Convergence in multimedia refers to:
A. Recording audio only
B. Merging of media and computing
C. Printing images
D. Deleting media files

24. Hypermedia is best described as:
A. A printed manual
B. Text-only media
C. A single video file
D. Multimedia with hyperlinks

25. MPEG is:
A. A storage device
B. An image editor
C. A font
D. A video compression family

26. Resolution refers to:
A. Font size
B. Number of pixels in an image
C. Colour name
D. File type

27. Which is an example of linear multimedia?
A. A game with choices
B. An interactive kiosk
C. A cinema presentation
D. A website with menus

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28. Morphing changes:
A. File format
B. Font size
C. One shape into another
D. Audio pitch

29. Path animation moves an object:
A. Randomly
B. Only vertically
C. Along a predefined route
D. Without motion

30. RGB is used mainly for:
A. Screen display
B. Audio recording
C. Video editing only
D. Printed output

PART D (40 MARKS)

QUESTION 1

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

QUESTION 2

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(5 marks)

QUESTION 3

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

(4 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

QUESTION 4

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

END OF QUESTION PAPER